Does the LLM Add Alpha? A Contamination-Free Forward Audit of a Deployed Multi-Agent LLM Trading System

Pradeep Somasundaram
Independent Researcher
aadhi1501@gmail.com

***Abstract*—Multi-agent large language model (LLM) trading systems are usually evaluated with historical backtests that fall inside the model's pre-training window and that credit the LLM for returns earned by the surrounding rules. We deploy a seven-node LangGraph agent (GPT-4o-mini, called by alias, with technical, fundamental, news and market-context inputs; a retrieval step is computed but never reaches the prompt) on a universe of U.S. large- and mid-cap equities and log 3,507 decisions in real time between January and September 2026, all after the model's knowledge cutoff. We assemble, from standard tools, a *Forward Audit Protocol* (FAP) that separates what the LLM contributes from what the pipeline contributes, and validate it by Monte Carlo on the real price panel: (i) a characteristic-matched placebo drawn from the same pre-screened feature space, (ii) block-bootstrap inference with an explicit minimum detectable effect, (iii) a calibration audit of the confidence the pipeline recorded, (iv) a specification–deployment audit with a counterfactual exit analysis, and (v) an operational audit of the production logs. At the agent's native intraday horizon, stated confidence averaged 0.82 while only 50.6% of BUY signals beat SPY (calibration gap 0.309), and the LLM's abnormal return over its matched placebo was -0.04% per trade (95% CI $[-0.12, +0.04]\%$), which rules out any edge above $+0.035\%$. Over ten days, 1,399 BUY signals earned $+1.71\%$, fully explained by market exposure: the LLM's abnormal return over its placebo was -0.13% (CI $[-1.21, +0.93]\%$), and the design could detect only edges above about 1.54% per ten days. The operational audit shows that 71.3% of logged decisions were exception fallbacks caused by an exhausted API quota, so the audited BUYs come from the 28.7% of decisions that reached the model. The Monte-Carlo checks show that the inference is well calibrated (false-positive rates of 3.4% at ten days and 6.0% intraday, against a nominal 5%); the intraday placebo is unbiased (+0.002 pp), but the ten-day placebo is biased by +0.27 pp for oversold picks. The result is a one-model case study with about five usable months, not a general verdict.**

***Index Terms*—Large language models, multi-agent systems, algorithmic trading, look-ahead bias, evaluation methodology, calibration, LangGraph, AI engineering.**

I. Introduction

LLM-based trading agents have progressed from sentiment classifiers [1], [2] to tool-using multi-agent systems that read news, fundamentals and price charts before issuing orders [3]–[6]. The reported gains, however, rest on two fragile foundations. First, backtests over historical periods overlap the LLM's pre-training corpus, so the model may "remember" outcomes [7]. Second, deployed agents are *pipelines*: a technical pre-screen chooses what the LLM sees, deterministic gates edit its output, and hand-coded rules set exits. A positive P&L therefore says little about whether the LLM itself adds information.

This paper asks a narrower but more useful question: *given everything else in the pipeline, does the LLM's choice add return?* We answer it with a forward, contamination-free audit of a system that ran in production for nine months. Our contributions are:

1) **A deployed multi-agent LLM trading system** with a four-stage funnel (pre-screen $\rightarrow$ seven-node LangGraph agent $\rightarrow$ deterministic gates $\rightarrow$ ATR exits), and a real-time log of 3,507 decisions on 472 tickers made after the model's knowledge cutoff.
2) **The Forward Audit Protocol (FAP)**, an audit assembled from standard tools (a characteristic-matched placebo and block-bootstrap inference) plus a specification–deployment consistency audit with a counterfactual exit analysis, a confidence-calibration audit and an operational audit of the production logs, *validated by Monte Carlo* on the real price panel for size, power and placebo bias (Section VI-C).
3) **Empirical evidence** that (a) the agent's selection alpha is statistically indistinguishable from zero at every horizon from intraday to ten days, with the detectable effect size stated for each; (b) the confidence recorded by the pipeline (after its gates) was over-confident and carried no rank information, most clearly at the native intraday horizon; (c) replaying the deployed intraday stop levels on the intended ten-day hold shows they would have discarded the market drift the selections captured; and (d) 71.3% of the logged decisions were fallbacks from an exhausted API quota, a failure that became visible only when the logs were audited.

The negative result is the point: an agent whose signals earned positive raw returns in a rising market can have no measurable LLM contribution once the right counterfactual is used, and a system that appears to run continuously can spend most of its decisions on a fallback path.

II. Related Work

LLMs in finance. Domain LLMs such as BloombergGPT [8] and FinGPT [2] showed that language models can extract financial signals; Lopez-Lira and Tang [1]

reported that ChatGPT headline scores predict next-day returns. **LLM trading agents** add memory (FinMem [3]), multimodal tools (FinAgent [4]) and role-playing analyst teams (TradingAgents [5]); see [6] for a survey. Most early work evaluates on historical windows that pre-date or overlap model training. Recent benchmarks close that gap by evaluating agents live or on post-cutoff windows: StockBench [9] finds that most LLM agents struggle to beat buy-and-hold, AI-Trader [10] reports that general intelligence does not ensure trading success, and Agent Market Arena [11] finds that agent architecture shapes behavior more than the underlying model. These studies compare agents with each other or with passive baselines; they do not isolate what the LLM's choice adds inside one deployed pipeline, which is the question we ask.

Evaluation pitfalls. Glasserman and Lin [7] documented look-ahead bias in GPT-based return prediction, Lopez-Lira et al. [12] showed that LLMs recall economic and financial data verbatim from before their cutoff and that instructions to respect historical boundaries do not stop it, and Gao et al. [13] give a statistical test for it that finds the bias disappears after the cutoff, which motivates forward evaluation. Kong et al. [14] review 164 papers and catalogue five recurring biases (look-ahead, survivorship, narrative, objective and cost) in financial LLM work, and Ye et al. [15] argue that reported LLM-agent alpha should not be treated as deployment evidence and propose a minimum reporting protocol. FAP is narrower and complementary: it supplies a placebo-based attribution, stated power, and an operational audit for a single deployed system. Data-snooping and multiple testing inflate backtest Sharpe ratios [16], [17]. Finance has long used characteristic-matched benchmarks [18] and event studies [19] to isolate skill from exposure; we bring these tools to LLM agents and add dependence-robust inference [20], [21].

LLM confidence. Neural networks are often miscalibrated [22]; LLMs partly know what they know [23], but verbalized confidence tends to be over-confident [24]. Trading agents nonetheless use it as a position gate, which we test directly.

III. System Architecture

Fig. 1 shows the production pipeline, which ran hourly during U.S. market hours on an AWS EC2 instance.

Stage 1 computes hourly RSI-14 [25], a volume ratio and short-term momentum for every constituent and forwards at most 50 candidates. **Stage 2** is a LangGraph [26] directed graph of seven nodes: price retrieval, technical features (RSI, MACD, ATR, trend, support/resistance), fundamentals (valuation, health and growth scores, analyst targets), news sentiment, market context (VIX, index and sector returns), retrieval-augmented context from a vector store of market patterns and risk guidelines [27], and a reasoning node. The reasoning node prompts GPT-4o-mini [28] (knowledge cutoff October 2023) with the technical, fundamental, news and market-context outputs (the retrieved context is computed but never inserted into the prompt; Appendix B) and returns a JSON object with

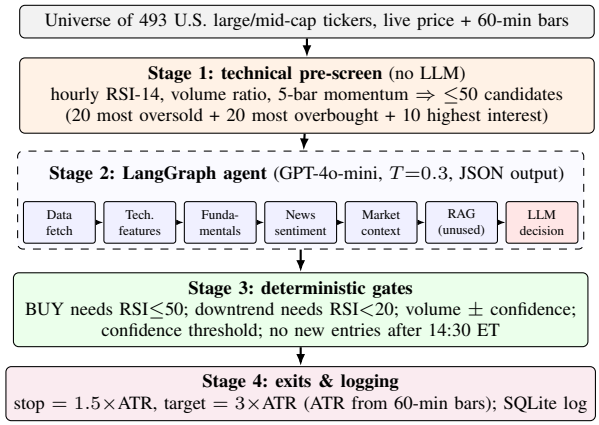


Fig. 1. Deployed four-stage pipeline. Only Stage 2 involves the LLM; FAP measures the incremental value of that stage's choice.

decision, confidence, target, stop and rationale. Deep analysis of the selected candidates ran in parallel worker threads. If the model call raises an exception, the agent returns HOLD with confidence 0.5 and logs the error; Section VI-H measures how often that happened. **Stage 3** applies deterministic gates that can cap confidence or relabel a decision (e.g., an LLM HOLD with RSI>80 becomes WATCH SELL). **Stage 4** sets ATR-based exits and writes each recommendation to a database with a UTC timestamp.

IV. The Forward Audit Protocol

A. Contamination-free signal set

Every decision is logged at generation time, after the model's cutoff, so no future information can reach the model. Hourly rescans repeat calls, so we keep the first signal per (ticker, day, decision). Signal i on ticker s_i issued on day t_i is entered at the close of t_i (robustness: next open) for multi-day horizons, and at the close of the 60-minute bar containing the timestamp for the intraday horizon. Both entries occur strictly after the signal.

B. Abnormal returns and event study

With $P_{s,t}$ the adjusted close and $\beta_{s,t}$ a 60-day rolling beta estimated before t, the beta-adjusted cumulative abnormal return is

$$\mathrm{CAR}_i(h) = \frac{P_{s_i,t_i+h}}{P_{s_i,t_i}} - 1 - \beta_{s_i,t_i}\Big(\frac{M_{t_i+h}}{M_{t_i}} - 1\Big), \quad (1)$$

where M is SPY. We trace $\overline{\mathrm{CAR}}(h)$ for $h = 0, \ldots, 10$ [19].

C. Characteristic-matched placebo

Comparing against SPY or random stocks conflates the LLM's choice with the pre-screen's tilt toward oversold, volatile names, which carries its own short-term reversal premium [29]. FAP instead builds, for each signal, a placebo of the K stocks that look most similar to the chosen one on

Algorithm 1 Characteristic-matched placebo

Input: signal (s_i, t_i); universe $\mathcal{U}$; LLM picks $\mathcal{S}_{t_i}$; features $x_{s,t} \in \mathbb{R}^d$

$\mathcal{C} \leftarrow \mathcal{U} \setminus \mathcal{S}_{t_i}$ with complete data on $[t_i, t_i + h]$

$z_s \leftarrow (x_{s,t_i} - \mu_{\mathcal{C}})/\sigma_{\mathcal{C}}$ {cross-sectional z-score}

$\mathcal{N}_K(i) \leftarrow K$ members of $\mathcal{C}$ minimizing $\|z_s - z_{s_i}\|_2$

Output: $\Delta_i(h)$ via Eq. (2)

the same date but were *not* chosen (Algorithm 1) [18]. The LLM's value-added is

$$\Delta_i(h) = \mathrm{CAR}_i(h) - \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \mathrm{CAR}_j(h). \qquad (2)$$

For multi-day horizons the features are daily RSI-14, 20-day momentum and ATR/price. For the intraday horizon they are the pre-screen's own features: hourly RSI-14, 5-bar momentum, hourly ATR/price and volume ratio, all measured at the last completed bar. We set K=10 and also report a random same-date control (K=20).

D. Dependence-robust inference

Signals on the same day share market shocks, and 10-day windows overlap. FAP resamples non-overlapping blocks of L=10 trading days (one day for intraday) with replacement [20], B=5,000 times, and reports percentile 95% CIs and two-sided bootstrap p-values. Portfolio alphas use Newey–West standard errors with 10 lags [21].

E. Specification–deployment consistency and exit decomposition

Agent pipelines have hidden technical debt [30]. FAP checks that four horizons agree: the prompt's horizon, the feature timeframe, the exit levels, and the holding period being evaluated. It then splits trade return into *selection* (hold to horizon) and *execution* (deployed exit minus hold). Execution is also replayed over a grid of stop/target multiples of *daily* ATR. Intrabar ordering is unknown, so the stop is assumed to fill first, and gaps fill at the open. All trades pay 10 bps round trip.

F. Calibration audit

We treat stated confidence c_i as a probability that the trade beats SPY and report the expected calibration error (ECE) over confidence bins [22], the Brier score against a constant (climatological) forecast, and the block-bootstrapped Spearman correlation between c_i and $\mathrm{CAR}_i(10)$.

V. Experimental Setup

The production log contains 3,507 decisions from 2026-01-09 to 2026-09-10. After de-duplication, 1,399 BUY signals on 393 tickers over 116 trading days have complete 10-day forward data (outcomes through 2026-09-24), with 12.1 signals per active day on average. We also evaluate 1,414 intraday BUY signals on 119 days and 1,308 WATCH SELL signals (the LLM said HOLD on names with RSI>80), which serve as an overbought reference set. The mean stated confidence of BUYs is 0.82. The scanner's universe is a fixed list of 493 U.S. large- and mid-cap stocks, broader than the S&P 500: of the 472 tickers in the log, 47 (mid-cap and recently listed names; none is an ETF) were not index members on 2026-01-01, and they account for 14.7% of the evaluated BUY signals. The placebo pool is the fixed list of 504 S&P 500 constituents on 2026-01-01 (499 with complete price data), reconstructed from Wikipedia's constituent list and a public index-change log. Non-members are evaluated as signals but never serve as placebo names. An earlier version of this analysis drew the pool from tickers that appeared in the log, which can peek ahead; Section VI-I shows that this mattered. Prices are split- and dividend-adjusted daily and 60-minute bars from Yahoo Finance. Over the same window SPY returned $+11.4\%$. Because of the API outage of Section VI-H, most signals come from April onward (1,186 of 1,399), so the usable window is about five months, not nine.

What is audited. A BUY in the log is the LLM's decision after the Stage-3 gates: Stage 3 changed the stated confidence of 40.9% of BUYs, and HOLD decisions were almost never stored (67 of 3,507 rows), so the candidates the LLM saw and rejected are not available. FAP therefore measures the value added by *the LLM plus its gates* against a placebo built from daily (multi-day) or pre-screen (intraday) features, which approximates but is not identical to the pre-screen's candidate set. Lineage checks out: 95.9% of the 1,493 database BUYs match a same-ticker LLM BUY in the production log, a median of 0.7 minutes earlier.

Statistical power. With bootstrap standard error SE of Δ, the minimum detectable effect at 5% two-sided significance and 80% power is $\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE} \approx 2.80\,\mathrm{SE}$ (Table III). A null result should be read as "no edge larger than the MDE", not as "no edge". This MDE is for the LLM-minus-placebo difference over all signals; comparisons of one selection rule with another (Table VII) are noisier and use fewer independent blocks per window, hence their larger minimum detectable differences. Section VI-C measures size and power directly instead of relying on this formula.

VI. Results

A. Selection: does the LLM pick better stocks?

Table I and Fig. 2 give the central result. BUY signals earned $+1.71\%$ over ten days (p=0.020), but after adjusting for market exposure the CAR is -0.04% and indistinguishable from zero. The difference from the matched placebo is -0.13% (CI $[-1.21, +0.93]\%$). The placebo itself is no better than random ($+0.23\%$), so the pre-screen features carried no premium in this sample either. We do not compare per-signal hit rates with the placebo's: a placebo built as a ten-stock average is positive more often than any single stock, so that comparison would not be like for like. The CAR difference above is the like-for-like measure.

Horizon robustness. Table II shows that the null result holds at every horizon, including the agent's native intraday horizon. It also holds with next-open entry (measured as

TABLE I
SELECTION ALPHA OF LLM BUY SIGNALS (10 TRADING DAYS, %)

Measure	N	Mean	95% CI	p
LLM BUY, raw return	1,399	$+1.71$	$[+0.18, +3.46]$	0.020
LLM BUY, excess over SPY	1,399	-0.10	$[-1.06, +0.97]$	0.838
LLM BUY, CAR	1,399	-0.04	$[-1.10, +1.04]$	0.942
Matched placebo, CAR	1,399	$+0.09$	$[-1.05, +1.23]$	0.840
Random same-date, CAR	1,399	-0.13	$[-1.06, +0.76]$	0.772
Overbought (WATCH SELL), CAR	1,308	$+0.31$	$[-0.60, +1.15]$	0.482
Δ LLM $-$ placebo	1,399	-0.13	$[-1.21, +0.93]$	0.803
Δ LLM $-$ random	1,399	$+0.09$	$[-1.05, +1.22]$	0.853

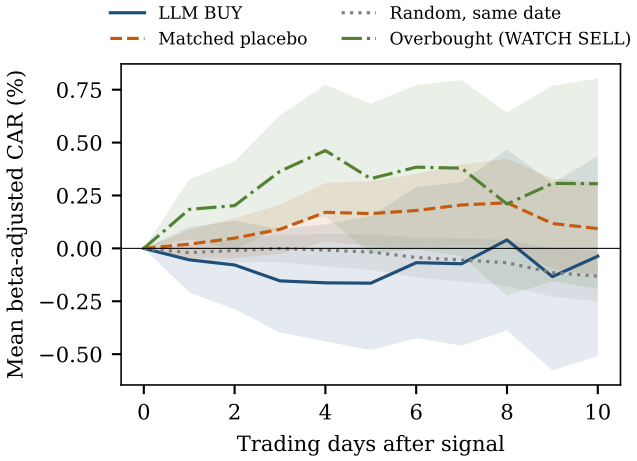


Fig. 2. Event study of beta-adjusted CAR after each signal. Bands are pointwise 95% intervals that ignore dependence; Table I reports block-bootstrap inference.

excess return over SPY, not over the placebo: -0.23%, CI $[-1.29, +0.93]$) and after dropping the most oversold decile (Table II). In the intraday case, 10 bps of costs turn a zero-alpha strategy into a significant loss of -0.16% per trade (CI $[-0.28, -0.04]\%$).

What the null can and cannot rule out. Only the intraday test is sharp: its interval excludes any edge above $+0.035\%$ per trade, below the 10 bps cost. At ten days the MDE is 1.54%; an edge of 0.5% per ten days, worth roughly 13% per year if it could be traded continuously, lies well inside the interval. The ten-day result therefore means "no edge larger than about 1.54%", not "no edge".

B. Do simple rules do as well?

A selection result is only meaningful next to trivial alternatives. On each of the 116 dates with LLM BUYs, four rules chose the same number of stocks from the fixed index pool using one column of the price panel and no model (Table IV). None is distinguishable from the LLM. The lowest-RSI rule is nominally best, but its interval spans zero as well, so with the power of Table III we can neither confirm nor exclude an edge of that size.

TABLE II
LLM VALUE-ADDED Δ OVER THE MATCHED PLACEBO BY HORIZON (%)

Horizon / variant	Δ	95% CI
Intraday (bar close $\rightarrow$ 16:00 ET)	-0.04	$[-0.12, +0.04]$
1 day	-0.07	$[-0.29, +0.14]$
3 days	-0.24	$[-0.91, +0.30]$
5 days	-0.33	$[-1.40, +0.53]$
10 days	-0.13	$[-1.19, +0.94]$
10 days, excl. most-oversold decile	-0.16	$[-1.16, +0.89]$

TABLE III
MINIMUM DETECTABLE EFFECT (5% TWO-SIDED, 80% POWER) FOR Δ OVER THE PLACEBO (%)

Horizon	Bootstrap SE	MDE	CI upper bound
Intraday	0.040	0.112	+0.035
1 day	0.110	0.31	+0.14
3 days	0.305	0.85	+0.30
5 days	0.491	1.38	+0.53
10 days	0.551	1.54	+0.94

C. Does the audit itself work?

We validated the inference on the real price panel. 500 fake LLMs each picked, on every date with real BUYs, the same number of stocks at random from the 50 lowest-RSI names (mimicking the pre-screen) and were compared with the identical matched placebo; by construction they have no skill. Two facts follow (Table V). First, once the mean is centred, the procedures do not over-reject: the percentile block bootstrap used in this paper wrongly finds an edge in 3.4% of fake LLMs (nominal 5%), a cluster-robust t test in 2.4% and a sign-flip permutation test in 1.4%, so 15 independent blocks suffice for size, and an injected edge of 1 pp is found 99.2% of the time (62.6% at 0.5 pp). This holds prices fixed and randomises only the picks, so it validates size and power against pick randomness, not against a different market path. Second, the placebo is biased for extreme picks: a fake LLM with no skill earns +0.27 pp over its matched placebo (SD 0.19 pp), so the right zero for an oversold picker is not zero, and the real LLM's -0.13% lies below 2% of the fake ones.

Table VI shows that this is design-dependent and not a finding. Under every placebo design the LLM's alpha stays within noise of zero (from -0.31 to $+0.23$ pp, every interval spanning zero: yes), the bias of no-skill pickers ranges from $+0.16$ to $+0.37$ pp, and once the placebo also matches on 5-day return, distance below the 20-day high and beta, the LLM does worse than only 35% of no-skill pickers. We therefore keep the conclusion at "no detectable edge" and do not claim negative selection skill.

The intraday result, which the abstract leads with, was validated the same way (119 day-blocks; Tables V and VI). The bootstrap's false-positive rate is 6.0% (cluster-robust t: 5.7%, sign-flip: 5.3%) at a nominal 5%, an injected edge of 0.08 pp is found about four times in five, and, unlike the ten-day placebo, the intraday placebo is essentially unbiased: no-skill pickers earn +0.002 pp (SD 0.028 pp). Across the six

TABLE IV

10-DAY CAR (%) OF THE 1,399 LLM BUY SIGNALS AND OF RULES PICKING THE SAME NUMBER OF STOCKS PER DATE; LAST COLUMNS: LLM MINUS RULE.

Strategy	Mean	95% CI	LLM − rule	95% CI
LLM BUY signals	-0.04	$[-1.06, +1.00]$	—	—
Whole universe	-0.07	$[-1.02, +0.85]$	$+0.04$	$[-1.14, +1.15]$
Top 20-day momentum	$+0.58$	$[-2.61, +3.70]$	-0.62	$[-3.57, +2.48]$
Lowest daily RSI	$+1.12$	$[-0.59, +2.90]$	-1.16	$[-3.12, +0.49]$
Lowest ATR%	$+0.08$	$[-0.74, +0.81]$	-0.12	$[-1.26, +1.11]$

TABLE V

MONTE-CARLO CHECK OF THE INFERENCE (%): SHARE OF NO-SKILL FAKE LLMS (500 AT TEN DAYS, 300 INTRADAY) FOR WHICH EACH PROCEDURE REPORTS AN EDGE, AFTER CENTRING, WHEN A TRUE EDGE δ IS ADDED; δ_0=0 IS THE FALSE-POSITIVE RATE (NOMINAL 5%). δ IN PP: TEN-DAY 0, 0.5, 1.0, 1.5, 2.0; INTRADAY 0, 0.03, 0.05, 0.08, 0.12.

Procedure	δ_0	δ_1	δ_2	δ_3	δ_4
Ten-day					
percentile block bootstrap	3.4	62.6	99.2	100.0	100.0
cluster-robust t	2.4	52.6	98.0	100.0	100.0
sign-flip permutation	1.4	46.2	97.6	100.0	100.0
Intraday					
percentile block bootstrap	6.0	19.0	47.0	79.3	98.7
cluster-robust t	5.7	18.3	45.7	79.0	98.7
sign-flip permutation	5.3	18.0	45.3	79.0	98.7

placebo designs the LLM's intraday alpha ranges from -0.049 to -0.029 pp (every interval spanning zero: yes), no-skill pickers earn -0.005 to $+0.008$ pp, and under the paper's own design the LLM does worse than 95% of them: suggestive, not significant. The abstract's claim that an intraday edge above $+0.035\%$ is ruled out therefore stands, with the simulation's own resolution (an empirical minimum detectable effect of about 0.079 pp) as its precision. The intraday standardisation uses all universe names valid at the bar rather than excluding the day's picks, a small approximation.

D. Consistency over time and a pre-registered forward test

Splitting the signal dates at 60% (1,140 signals in 9 independent ten-day blocks, then 259 in 7) and evaluating six fixed hypotheses in each window (Table VII) finds no detectable effect in either. The lowest-RSI rule's advantage over the universe is positive in both windows, yet its interval always spans zero; confidence gating and confidence ranking are null in both. Because signals cluster in time, the effective sample is the number of independent blocks (15), so the smallest detectable difference between rules is about 2.4 percentage points. The lowest-RSI hypothesis (H3) and H5 were suggested by development results, so these windows are exploratory for them; only forward data can confirm either. H6 is the primary test of selection skill: the ten-day alpha over the placebo minus the placebo's measured bias (Section VI-C), with the bias estimated from seeded no-skill fake pickers on the same dates and the whole procedure fixed before any forward outcome existed. We therefore froze the six hypotheses, thresholds and a minimum sample (300 matured signals and 20 independent blocks) in a committed pre-registration on 2026-09-25, before any outcome of the forward window existed. Below that sample the test reports "insufficient data" rather than findings. As of 2026-09-25, 0 of the 300 matured signals and 0 of the 20 independent blocks required by the pre-registration had matured, so no forward result is reported. Two further exploratory analyses, a purged walk-forward meta-model and an input ablation, are in Appendices A and B; neither changes the conclusion.

TABLE VI

SENSITIVITY TO THE PLACEBO'S DESIGN (PP). LLM: ALPHA OVER THE PLACEBO WITH 95% CI; NO-SKILL: THE SAME QUANTITY FOR RANDOM OVERSOLD PICKERS (THE PLACEBO'S BIAS); LLM − BIAS; LAST COLUMN: SHARE (%) OF NO-SKILL PICKERS THE LLM DOES WORSE THAN.

Placebo design	LLM	No-skill	LLM − bias	Worse than
Ten-day (daily-feature placebo)				
base (K=10)	-0.13 $[-1.17, +0.94]$	$+0.26$	-0.39	96
K=5	-0.08 $[-0.98, +0.93]$	$+0.16$	-0.24	88
K=20	-0.14 $[-1.28, +0.97]$	$+0.37$	-0.50	100
richer features	$+0.23$ $[-0.73, +1.22]$	$+0.16$	$+0.07$	35
same sector	-0.28 $[-1.24, +0.88]$	$+0.34$	-0.62	100
index-only signals	-0.31 $[-1.30, +0.69]$	$+0.24$	-0.55	99
Intraday (hourly-feature placebo)				
base (K=10)	-0.044 $[-0.121, +0.034]$	$+0.002$	-0.046	95
K=5	-0.049 $[-0.126, +0.026]$	-0.002	-0.047	95
K=20	-0.033 $[-0.114, +0.044]$	$+0.005$	-0.039	96
richer features	-0.042 $[-0.121, +0.037]$	-0.005	-0.038	90
same sector	-0.029 $[-0.107, +0.046]$	$+0.008$	-0.037	94
index-only signals	-0.035 $[-0.115, +0.043]$	$+0.004$	-0.039	93

TABLE VII

FIXED HYPOTHESES IN THE EARLY AND LATE DEVELOPMENT WINDOWS: 10-DAY CAR DIFFERENCE IN % (RANK CORRELATION FOR H2), WITH 95% CI.

Hypothesis	Early window	Late window
H1: conf. ≥ 0.85 vs. all	-0.09 $[-0.72, +0.76]$	$+0.96$ $[-1.25, +2.31]$
H2: rank corr. conf.–CAR	-0.008 $[-0.059, +0.050]$	$+0.073$ $[-0.115, +0.179]$
H3: low-RSI vs. universe	$+0.78$ $[-0.72, +2.51]$	$+3.01$ $[-1.67, +5.34]$
H4: LLM vs. low-RSI	-0.54 $[-1.74, +0.50]$	-3.90 $[-7.28, +2.99]$
H5: lowest-RSI third vs. all (placebo alpha)	-0.10 $[-0.86, +0.56]$	-0.78 $[-1.48, +0.48]$
H6: bias-corrected alpha (primary)	-0.11 $[-1.19, +1.05]$	-1.65 $[-2.87, +1.11]$

E. Specification–deployment mismatch: a counterfactual exit analysis

The project was designed for 5–10-day swing trades, and the codebase contains a swing prompt that asks for stops of 2–3× *daily* ATR. The consistency audit found that the production scanner instead called the *intraday* prompt on 60-minute bars, with stops at 1.5× *hourly* ATR, and closed every position the same day: of 1,483 BUYs with a recorded exit, 89.1% were closed at the end of day and the other 161 by the intraday stop. No position was ever held for ten days with these levels, so the analysis below is a *counterfactual*: it asks what would have happened had the deployed stop and target been attached to the ten-day hold the project was designed for. Measured against daily ATR, that stop is only 0.49× (median 1.87% below entry). Table VIII shows the replay: 65.5% of positions would have been stopped out, the win rate would have fallen from 53.2% to 33.9%, and mean net return from

TABLE VIII
COUNTERFACTUAL EXIT DECOMPOSITION FOR 10-DAY BUY POSITIONS (NET OF 10 BPS, %)

Exit policy	Mean	95% CI	Win	Stop	Target
Hold 10 days	$+1.61$	$[+0.09, +3.36]$	53.2	–	–
Deployed levels	-0.06	$[-0.31, +0.17]$	33.9	65.5	32.9
Counterfactual effect	-1.67	$[-3.23, -0.27]$	$p = 0.008$		

TABLE IX
CALIBRATION OF LLM CONFIDENCE (BUY SIGNALS, 10 DAYS)

Confidence bin	N	$\bar{c}$	P(beat SPY)	P(target 1st)	Δ (%)
$[0.0, 0.7)$	48	0.59	43.8	25.0	$+2.07$
$[0.7, 0.8)$	330	0.75	42.7	31.2	-0.23
$[0.8, 0.85)$	671	0.82	45.5	34.0	-0.32
$[0.85, 0.9)$	332	0.90	46.1	32.5	$+0.10$
$[0.9, 1.0]$	18	0.96	44.4	50.0	-1.36

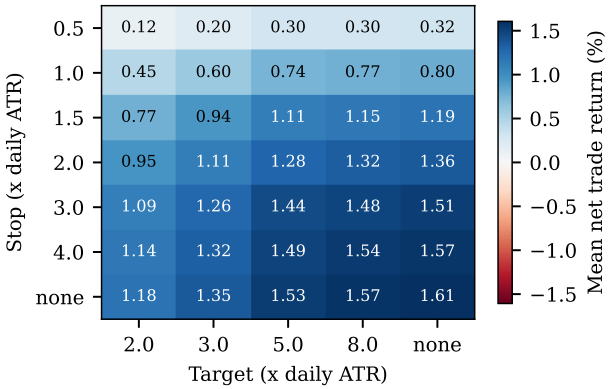


Fig. 3. Mean net trade return (%) for stop/target multiples of *daily* ATR over a 10-day horizon. The deployed stop lies near the 0.5 row.

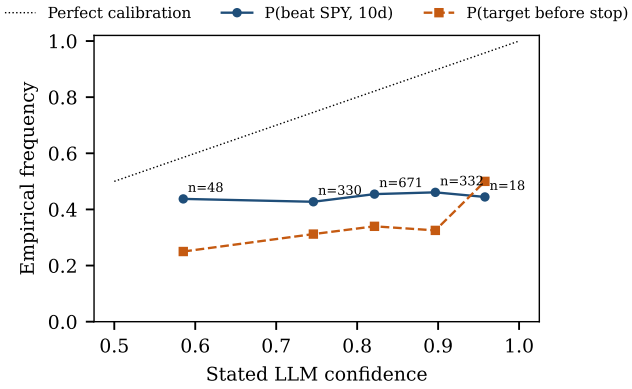


Fig. 4. Reliability diagram. Empirical success is flat across stated confidence.

$+1.61\%$ to -0.06%, a difference of -1.67% per trade (CI $[-3.23, -0.27]\%$). Fig. 3 shows that across the grid, returns rise steadily as stops widen, and no stop or target beats simply holding ($+1.61\%$); the tightest setting earns $+0.32\%$. Because the replay applies an intraday rule to a horizon it was not built for, it quantifies a configuration error, not a realised loss. Its realised counterpart is the intraday analysis, where the deployed levels stopped 10.9% of trades (161 of 1,483 in production) and changed mean return by only -0.00% (CI $[-0.04, +0.04]\%$).

F. Calibration of the recorded confidence

Native intraday horizon. At the horizon the prompt was written for, stated confidence averaged 0.82 while 50.6% of BUYs beat SPY the same day, a calibration gap of 0.309. **Ten-day horizon.** Graded against a question the scores were not written for, confidence was concentrated at 0.82 while only 44.9% of BUYs beat SPY over ten days (Table IX, Fig. 4). The ECE is 0.366 and the Brier score is 0.386, worse than the 0.247 of a constant forecast. The rank correlation with CAR is $\rho = +0.008$ (CI $[-0.045, +0.064]$), so confidence did not even order trades correctly. Using it to gate trades, as Stage 3 does, is therefore unfounded. Because Stage 3 altered the confidence of 40.9% of BUYs, the audited value is the pipeline's, not the model's alone, so the resemblance to evidence that LLMs verbalize over-confidence [24] is suggestive only; raw model confidence is logged from September 2026 on.

G. Portfolio view

Table X and Fig. 5 simulate a signal-weighted portfolio that holds each BUY for ten days, with the placebo and a mechanical "daily RSI<25" rule built the same way. This simulation lets same-ticker signals stack and imposes no capital constraint, so we report it as context rather than evidence. The LLM portfolio's Sharpe ratio exceeds SPY's (1.70 vs. 1.21), but its beta is 0.53 and its HAC alpha is not significant (t=0.94). The long LLM / short placebo spread earned $+6.7\%$ (t=0.68). The portfolio weights days equally, whereas Table I weights signals equally, so busy days count less here. The gap between the two views is itself a warning: a headline Sharpe ratio over nine months is well within the range that selection and multiple testing can produce [16], [17].

H. Operational audit of the production log

The production log covers 842 hourly scans (median duration 1.6 minutes, about 38 candidates per scan) and contains 67,810 decisions. Only 19,467 (28.7%) came from the model. The other 48,343 are the agent's exception fallback: 48,340 were caused by an exhausted OpenAI quota ("You exceeded your current quota") on 28 distinct days between 2026-02-06 and 2026-05-18, and only 2 by unparsable model output. Real decisions plus fallbacks reconcile with the number of decision lines in the log. Table XI shows that the outage covers almost all of February, which is why the database holds no February signals, and most of March; from April every decision reached the model. The API returned 286,876 HTTP 429 responses (including automatic retries) against 36,864 successful chat completions. For decisions that reached the model, the median

TABLE X
PORTFOLIO SIMULATION, 2026-01-09 TO 2026-09-24 (10 BPS COSTS)

Strategy	Ret.%	Vol.%	Sharpe	MDD%	α%/yr (t)	β
LLM agent	+17.4	14.0	1.70	−7.7	+15.2 (0.94)	0.53
Matched placebo	+9.4	13.8	1.00	−7.8	+3.1 (0.26)	0.66
RSI<25 rule	+8.8	19.8	0.71	−13.5	+1.9 (0.09)	0.74
SPY	+11.4	13.4	1.21	−8.9	–	1.00
LLM − placebo	+6.7	12.3	0.82	−7.1	+10.0 (0.68)	–

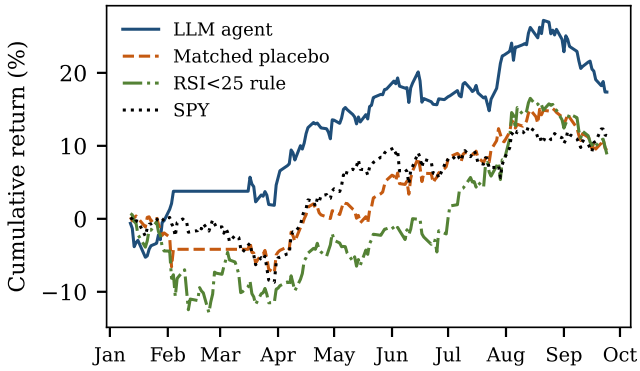


Fig. 5. Cumulative return of signal-weighted 10-day portfolios and SPY.

time from the start of a ticker's analysis to its decision was 9.4 s (95th percentile 19.3 s) with five parallel workers.

Signals from the outage rest on the few candidates that did reach the model, so we split the ten-day result. Signals before April (n = 213) show a Δ of $+2.02\%$ (CI $[+0.37, +4.26]\%$, p = 0.036) with an MDE of 4.00%, resting on very few independent blocks; signals from April onward, when every decision reached the model (n = 1,186), show -0.52% (CI $[-1.50, +0.40]\%$, MDE 1.35%). We do not interpret the pre-April estimate: it is the only nominally significant result in the study, comes from the period with the least trustworthy sampling, and would not survive a Holm correction across the subset tests reported here.

What the model said. Stored rationales are the model's key factors. 95.8% of BUYs cite "RSI is oversold" (median stored RSI of BUYs: 16.9), none of those contradict the stored RSI, and the three most common factor strings account for 38.0% of all cited factors among 333 distinct strings. The model's choice is largely a restatement of the pre-screen's oversold flag, which is consistent with the null selection result. The database stopped storing news and fundamentals scores after January, so we cannot audit whether those inputs influenced decisions; logging them is the first change we recommend.

I. Exploratory regime analysis

Table XII splits Δ by VIX on the signal date and by SPY trend. Only one estimate is nominally significant: the LLM trailed its placebo when VIX was below 16 (it also survives a Holm correction across the five tests, though these bootstrap p-values rest on few independent blocks). The positive estimate for VIX above 20 is not distinguishable from zero. It is also

TABLE XI
SHARE OF LOGGED DECISIONS THAT REACHED THE LLM, BY MONTH

Month	Decisions	LLM output	Fallback
2026-01	636	100%	0%
2026-02	31,666	7%	93%
2026-03	21,168	11%	89%
2026-04	4,019	100%	0%
2026-05	3,860	100%	0%
2026-06	3,911	100%	0%
2026-07	722	100%	0%
2026-08	1,183	100%	0%
2026-09	645	100%	0%

TABLE XII
LLM VALUE-ADDED BY MARKET REGIME (10 DAYS, %)

Regime	N	CAR	Δ	95% CI	p
VIX<16	142	−0.17	−0.79	$[-1.46, -0.25]$	0.006
16-20	925	−0.34	−0.58	$[-1.74, +0.51]$	0.308
VIX>20	332	+0.86	+1.41	$[-1.23, +2.68]$	0.243
SPY>SMA50	1087	−0.31	−0.62	$[-1.58, +0.36]$	0.212
SPY<SMA50	312	+0.90	+1.56	$[-0.45, +2.82]$	0.098

the estimate that moved most when we fixed the placebo pool: with a pool drawn from tickers that appeared in the log, the same split gave +1.90% (CI [+0.27, +3.31]), which looked like value under elevated volatility and would have been a headline. Fixing the pool to the constituents on the first study day removed it, a concrete illustration of how sensitive regime splits are to the placebo universe. Because the splits are post hoc and the high-VIX sample covers only a few episodes, the table should be read descriptively, and this result is kept out of the abstract.

VII. DISCUSSION

Lessons for AI engineering. (1) *Attribute before you celebrate.* The agent's portfolio beat SPY on Sharpe (1.70 vs. 1.21) and its placebo (1.00), yet neither its alpha (t=0.94) nor its long–short spread over the placebo (t=0.68) is distinguishable from zero, and the placebo's own Sharpe moved from 1.37 to 1.00 when the pool was fixed. P&L is a property of the whole pipeline, and the LLM's contribution must be measured against a counterfactual that removes only the LLM's choice. (2) *Specification drift is a first-class failure mode.* A single wrong flag sent the intraday prompt into production for a swing strategy. Every component then behaved "correctly" in isolation while a counterfactual replay shows it would have lost -1.67% per trade on the intended horizon. Consistency checks between prompt horizon, feature timeframe, exit rules and evaluation horizon belong in the test suite. (3) *Do not gate on the recorded confidence* unless it has been calibrated on forward data, and log the model's raw confidence separately from the gated one. (4) *Log the failure path.* Seven in ten decisions came from a silent exception fallback that emitted a plausible-looking HOLD; fallbacks need their own label, counter and alert.

Threats to validity. The sample spans nine months, 116 signal days and a single model (GPT-4o-mini), so power is limited (Table III); a 2024-vintage model says nothing about frontier models. The model was called by the alias gpt-4o-mini and the snapshot served during the study was not logged (the system now logs it; on 2026-09-25 the alias resolved to gpt-4o-mini-2024-07-18), so a silent snapshot change cannot be excluded. Decisions come from a stochastic process (temperature 0.3): in a pilot replay of 38 candidates two identical runs disagreed on 6.6% of decisions, and every result here is one realisation of it. The matched placebo is biased for oversold picks (Section VI-C), and the pre-registration, though committed to the project repository, carries only the author's own timestamps. The February gap is not random downtime but the quota outage of Table XI, and signals from that period rest on the small share of candidates that reached the model. The placebo pool is a fixed list of index constituents reconstructed from two public web sources that we did not independently audit, and it excludes the 47 non-index tickers the scanner also traded, so matching is among index names only. After the study period the scan list was aligned to the current index (2026-09-25), so forward results come from a slightly different set of stocks; that change is logged with the pre-registration. Stage-3 gates edit the LLM's output, HOLDs were not stored, and the multi-day placebo uses daily rather than pre-screen features, so the audit measures the LLM plus its gates against an approximation of its candidate set. The ten-day exit analysis is a counterfactual, since no position was held ten days. We assume 10 bps costs, no market impact and error-free Yahoo Finance bars; token counts and per-decision API cost were not logged. FAP does not prove that LLM agents cannot add value, only that this agent did not do so measurably.

VIII. Conclusion

We ran a multi-agent LLM trading system in production after the model's knowledge cutoff and audited its 3,507 logged decisions with the Forward Audit Protocol. The LLM's stock selection added no statistically detectable return over characteristic-matched alternatives at any horizon (minimum detectable effects in Table III). A counterfactual replay shows that the deployed intraday stop levels would have turned the captured market drift into a flat result, the pipeline's recorded confidence was uninformative, and 71.3% of the logged decisions were quota-exhaustion fallbacks. FAP (contamination-free logging, matched placebos, block-bootstrap inference with stated power, consistency, calibration and operational audits) is cheap to adopt, but it has been validated on one system only and its placebo needs a bias correction for extreme picks. We offer it as a starting point for auditing LLM agents, not as a settled standard. A purged walk-forward meta-model found no predictor of which calls work, and an input ablation showed that only the technical block moves decisions beyond the model's own randomness. A pre-registered forward test (six fixed hypotheses; minimum sample of 300 signals in 20 independent blocks) is running and will be reported once it matures. Future work will compare frontier models under the same protocol and re-test regime effects out of sample.

Data and Code Availability

The production decision log (model rationales included, news text excluded), the frozen price panel behind every number, the scripts that generate each table and macro in this paper, the pre-registration and its addenda, and the deployment code are in the project repository (github.com/PradeepSomasundaram1512/stock, directories ai-stock-agent and paper). Two runs of the analysis pipeline are byte-identical. The pre-registration carries the repository's commit timestamps only.

Appendix A
Can anything predict which calls work?

We tried to learn which BUYs beat their matched placebo, using purged walk-forward validation (expanding window, 10-day purge and 5-day embargo; 645 out-of-fold signals in 9 independent blocks) with two model families fixed in advance. Feature set A is price context only (RSI, momentum, distance below the 20-day high, ATR, beta, VIX, market trend, days to earnings); set B adds what the LLM said (stated confidence and trend label). Fundamentals, news and market fields could not be used, because the historical log recorded them for only about 10 of 1,493 BUYs. No model predicts the outcome out of sample (Table XIII): AUC is at or below chance, every selective-alpha interval spans zero, and the LLM's outputs add nothing measurable in either model family. The logistic models score below chance, meaning relations learned in earlier months reversed later. Of 9 tests, only the lowest-RSI third of BUYs underperforming survives a Holm correction, and only marginally (adjusted p=0.05); it was not hypothesised in advance, and it is not stable across windows (-0.10% early, -0.78% late, Table VII), so we pre-registered it as H5 for the forward test rather than reporting it as a finding. As checks on the pipeline, its out-of-fold AUC stays near 0.5 on shuffled noise and a planted signal is recovered (AUC above 0.7), so the null is not an artifact of the procedure. The minimum detectable selective alpha is about 2.4 percentage points. The analysis is exploratory: the target is a noisy placebo-relative alpha, the model families, the 30% cut and the features were chosen by the analyst, and Holm covers only the tests shown. A below-chance AUC for the logistic models more likely reflects instability across periods than a stable, reversed relation.

Appendix B
Which inputs move the decisions?

We replayed the exact logged inputs of 38 candidates from one scan through the production prompt, three times unchanged and once with each input group replaced by neutral text (Table XIV). Two identical runs disagree on $100 - 93.4\%$ of decisions, so the model's own randomness sets the bar. Removing the technical block changes decisions beyond it, after which the model buys far more often. This survives a

TABLE XIII

PURGED WALK-FORWARD META-LABELING. SELECTIVE ALPHA: MEAN ALPHA (%) OF THE TOP 30% BY PREDICTED SCORE MINUS ALL SIGNALS, 95% CI; LAST COLUMN: WHAT ADDING THE LLM'S OUTPUTS CONTRIBUTES OVER THE SAME MODEL WITHOUT THEM.

Model	AUC	Selective α	LLM adds
Logistic, price context	0.437	-1.33 $[-3.28, +0.42]$	—
Logistic, + LLM outputs	0.431	-1.51 $[-3.47, +0.05]$	-0.18 $[-0.58, +0.41]$
Boosted trees, price context	0.514	$+0.17$ $[-1.57, +1.90]$	—
Boosted trees, + LLM outputs	0.511	$+0.66$ $[-1.80, +3.02]$	$+0.49$ $[-0.27, +1.19]$
Top 30% by LLM confidence	—	-0.22 $[-0.93, +0.37]$	—
Lowest 30% by RSI	—	-0.77 $[-2.15, -0.19]$	—

TABLE XIV

INPUT ABLATION (38 CANDIDATES, ONE SCAN). AGREEMENT WITH AN UNCHANGED CONTROL RUN (%, 95% CI; TWO IDENTICAL RUNS AGREE 93.4%); BUY RATE (%; CONTROL 13%); DECISIONS FLIPPED TO / FROM BUY.

Variant	Agreement	BUY rate	To / from BUY
No news	87 [76,97]	3	0 / 4
No news (other wording)	87 [76,97]	8	1 / 3
No fundamentals	87 [76,97]	24	4 / 0
No market context	87 [76,97]	18	3 / 1
No technicals	74 [61,87]	37	9 / 0
No technicals (other wording)	58 [42,74]	47	13 / 0
Technicals only	84 [71,95]	5	1 / 4

second, differently worded replacement (58% agreement, BUY rate 47%), so it is not an artifact of the wording, although its size depends on it. News, fundamentals and market context each stay within noise at this sample size under every wording tried: removing news lowered the BUY rate (3% and 8% under the two wordings, control 13%) and removing fundamentals raised it, but these directions rest on a few flipped decisions and are not established. The retrieval-augmented context is not an ablation group: it is computed but never inserted into the prompt, so it cannot affect any decision. This is a one-scan pilot and a replacement string is never fully information-free. A weekly job now replays a sample and will report whether removing an input changes which BUYs earn returns once ten-day outcomes mature.

REFERENCES

[1] A. Lopez-Lira and Y. Tang, "Can ChatGPT forecast stock price movements? Return predictability and large language models," *arXiv preprint arXiv:2304.07619*, 2023.

[2] H. Yang, X.-Y. Liu, and C. D. Wang, "FinGPT: Open-source financial large language models," *arXiv preprint arXiv:2306.06031*, 2023.

[3] Y. Yu, H. Li, Z. Chen, Y. Jiang, Y. Li, D. Zhang, R. Liu, J. W. Suchow, and K. Khashanah, "FinMem: A performance-enhanced LLM trading agent with layered memory and character design," *arXiv preprint arXiv:2311.13743*, 2023.

[4] W. Zhang *et al.*, "A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist," in *Proc. 30th ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, 2024.

[5] Y. Xiao, E. Sun, D. Luo, and W. Wang, "TradingAgents: Multi-agents LLM financial trading framework," *arXiv preprint arXiv:2412.20138*, 2024.

[6] H. Ding, Y. Li, J. Wang, and H. Chen, "Large language model agent in financial trading: A survey," *arXiv preprint arXiv:2408.06361*, 2024.

[7] P. Glasserman and C. Lin, "Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis," *arXiv preprint arXiv:2309.17322*, 2023.

[8] S. Wu, O. Irsoy, S. Lu, V. Dabravolski, M. Dredze, S. Gehrmann, P. Kambadur, D. Rosenberg, and G. Mann, "BloombergGPT: A large language model for finance," *arXiv preprint arXiv:2303.17564*, 2023.

[9] Y. Chen, Z. Yao, Y. Liu, A. Xin, J. Ye, J. Yu, L. Hou, and J. Li, "StockBench: Can LLM agents trade stocks profitably in real-world markets?" *arXiv preprint arXiv:2510.02209*, 2025.

[10] T. Fan, Y. Yang, Y. Jiang, Y. Zhang, Y. Chen, and C. Huang, "AI-Trader: Benchmarking autonomous agents in real-time financial markets," *arXiv preprint arXiv:2512.10971*, 2025.

[11] L. Qian *et al.*, "When agents trade: Live multi-market trading benchmark for LLM agents," *arXiv preprint arXiv:2510.11695*, 2025.

[12] A. Lopez-Lira, Y. Tang, and M. Zhu, "The memorization problem: Can we trust LLMs' economic forecasts?" *arXiv preprint arXiv:2504.14765*, 2025.

[13] Z. Gao, W. Jiang, and Y. Yan, "Detecting lookahead bias in LLM forecasts," *arXiv preprint arXiv:2512.23847*, 2025.

[14] Y. Kong *et al.*, "Evaluating LLMs in finance requires explicit bias consideration," *arXiv preprint arXiv:2602.14233*, 2026.

[15] Y. Ye *et al.*, "The alpha illusion: Reported alpha from LLM trading agents should not be treated as deployment evidence," *arXiv preprint arXiv:2605.16895*, 2026.

[16] C. R. Harvey, Y. Liu, and H. Zhu, "…and the cross-section of expected returns," *Review of Financial Studies*, vol. 29, no. 1, pp. 5–68, 2016.

[17] D. H. Bailey and M. López de Prado, "The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality," *Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014.

[18] K. Daniel, M. Grinblatt, S. Titman, and R. Wermers, "Measuring mutual fund performance with characteristic-based benchmarks," *Journal of Finance*, vol. 52, no. 3, pp. 1035–1058, 1997.

[19] S. J. Brown and J. B. Warner, "Using daily stock returns: The case of event studies," *Journal of Financial Economics*, vol. 14, no. 1, pp. 3–31, 1985.

[20] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[21] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.

[22] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.

[23] S. Kadavath *et al.*, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[24] M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi, "Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.

[25] J. W. Wilder, *New Concepts in Technical Trading Systems*. Greensboro, NC: Trend Research, 1978.

[26] LangChain, Inc., "LangGraph: Build resilient language agents as graphs," https://github.com/langchain-ai/langgraph, accessed: Sep. 2026.

[27] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[28] OpenAI, "GPT-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.

[29] N. Jegadeesh, "Evidence of predictable behavior of security returns," *Journal of Finance*, vol. 45, no. 3, pp. 881–898, 1990.

[30] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison, "Hidden technical debt in machine learning systems," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 28, pp. 2503–2511, 2015.